TECHNICAL REPORT

Brick Kilns and Schoolchildren in the National Capital Region

2023–24 and 2025–26 firing seasons

Author	Rashmi Sinha
Discipline	GeoAI / Remote Sensing

Study area	National Capital Region, India (NCRPB definition)
Extent	55,016 km^2, 25 districts in 4 states
Coordinate system	EPSG:32643 — WGS 84 / UTM zone 43N
Imagery	Sentinel-2 surface reflectance, 10 m
Method	Oriented object detection (YOLO-OBB), SWIR firing verification
Software	Python 3.14, PyTorch, Ultralytics, QGIS 3.40.6, Earth Engine
Report date	October 13, 2026
Status	**Complete — 13 October 2026**

Written as a living document during the work: decisions, rejected methods and checks are recorded where they happened. Uncertainty labels used throughout: *verified*, *plausible*, *assumed*.

Contents

Executive summary **2**

1 Introduction **2**
1.1 Hypotheses . . . 3

2 Study area **3**
2.1 Boundary construction and verification . . . 4

3 Data sources **5**

4 Methodology **6**
4.1 Training data . . . 6
4.2 Image processing . . . 6
4.3 Spatially separated training and test sets . . . 6
4.4 NCR imagery . . . 7
4.5 Reference kiln points . . . 7
4.6 Children resident near kilns . . . 7
4.7 Change between seasons . . . 7

5 Results **8**
5.1 Wall-to-wall NCR scan . . . 8
5.2 Open question: not everything that burns is a kiln . . . 9
5.3 Schools and children within 800 m of a kiln . . . 10
5.4 Which kilns stopped firing, 2023–24 to 2025–26 . . . 12
5.5 H3: are kilns near schools closed more often? . . . 13
5.6 Can Sentinel-2 see kilns firing? (Phase 3 go/no-go) . . . 13
5.7 GRAP: do brick kilns have to stop? . . . 14
5.8 The firing calendar, November 2023 – June 2026 (step 22) . . . 14
5.9 Do kilns keep firing during air-quality emergencies? (step 23) . . . 15
5.10 How much smoke, and how much of it next to schools? (step 24) . . . 16
5.11 Rejected method: fine-tuning on this study's imagery . . . 18

6 Accuracy assessment **19**
6.1 School location accuracy . . . 19
6.2 Visual audit of 25 random kilns (measured precision) . . . 19
6.3 Spot check: a kiln not visible on the basemap . . . 20
6.4 Independent check of the exposure code (QGIS) . . . 21
6.5 Detector on held-out NCR patches . . . 21

7 Limitations **22**

A Processing parameters **24**

B Glossary **24**

C Decision log **25**

Executive summary

Question. The Central Pollution Control Board (CPCB) requires brick kilns to stand at least 800 m from schools and homes. How many children in the National Capital Region (NCR) live and go to school inside that distance? Which districts are worst affected? Is enforcement closing the kilns nearest to schools?

Approach. An AI object detector (YOLO11-OBB), trained on the SentinelKilnDB benchmark away from the NCR, was run over all 55,016 km^2 of the region on Sentinel-2 imagery. Detections were kept only if they passed four checks: land cover (brick-yard ground), firing (shortwave-infrared heat in the 2023–24 season), not a non-kiln heat source (airports, power plants, factories, year-round heat), and not a duplicate. The result was joined to 36,738 location-checked UDISE schools and WorldPop counts of children aged 5–14. Delhi NCT is excluded because kilns are not permitted there.

Main findings (2023–24 season, NCR excluding Delhi):

1. **About 605,000 children aged 5–14 (7.3%) live within 800 m of a firing brick kiln**, and **1,708 schools** stand inside that distance.

2. **Baghpat is the worst-affected district: one child in three (32.8%)** lives within 800 m of a kiln. It also has the most kiln smoke near schools (about 300 t $PM_{2.5}$ a season, low case).

3. **More than a third of kilns (34.6% of 3,256) have a school within 800 m**, so siting violations near schools are common (H1 confirmed).

4. **Enforcement does not target schools.** 13.9% of kilns stopped firing by 2025–26. The rate near schools (14.6%) is not clearly different from elsewhere (13.3%; $p = 0.31$, H3 confirmed). 962 kilns near schools were still firing in 2025–26.

5. **Kilns keep firing on air-quality emergency days**, because the GRAP emergency schedules do not order them to close. Whether they cut back on those days could not be measured with Sentinel-2.

6. **Kilns emit roughly 4,700–12,300 t of $PM_{2.5}$ a season**; about a third comes from kilns near schools. This is an estimate built on assumed production, not a measurement. The school with the most kilns nearby, JHS Rampur Kheri (Shamli), has 12 within 800 m.

How far to trust it. A visual audit of 25 random kilns on dated Google Earth imagery measured **precision 0.91** (95% CI 0.73–0.98). An independent re-count in QGIS reproduced the exposure code's numbers exactly. The firing calendar matches the known season (relighting in Oct–Nov, a peak in Feb–May, shutdown in the monsoon). Counts are *at least* figures: the detector misses some kilns (recall about 0.62 on this imagery). Children are counted where they *live* (modelled population); school enrolment was not available.

What should happen next. (1) Check the 962 kilns still firing within 800 m of a school against their consent-to-operate records, starting with Baghpat, Shamli and Muzaffarnagar. (2) Consider adding brick kilns to the GRAP Stage III/IV closure list, as stone crushers and mines already are. (3) Put low-cost $PM_{2.5}$ monitors at the most-affected schools (Rampur Kheri, Matiyala, Biharipur) to measure actual exposure.

1 Introduction

Brick kilns are estimated to contribute 8–14% of ambient air pollution in the Indo-Gangetic Plain [1]. CPCB rules require a kiln to stand at least 0.8 km from habitation, schools and hospitals, 1 km from another kiln and 500 m from a river [4]. Compliance is hard to check on the ground across thousands of kilns.

Automated kiln detection and compliance checking has been demonstrated [2]. This study adds two things: it counts the *children* affected, not only the schools, and it measures change between two seasons using one consistent detector.

1.1 Hypotheses

Recorded on 27 September 2026, before any kiln or school data was analysed. The predicted values are guesses, not derived from data.

	Hypothesis	Prediction	Rejected if
H1	Siting violations near schools are common	$\geq 25\%$ of NCR kilns have a school within 800 m	$< 10\%$
H2	Exposure is concentrated	Top 3 districts hold $> 50\%$ of exposed children, likely western UP	Spread evenly
H3	Enforcement does not target schools	Closure rate near schools $\approx$ closure rate elsewhere	Near-school kilns close clearly more often

2 Study area

The National Capital Region as defined by the NCR Planning Board: Delhi NCT, 14 districts of Haryana, 8 of Uttar Pradesh and 2 of Rajasthan (Figure 1). It is the jurisdiction of the Commission for Air Quality Management, the body that issues directions on kilns in the region.

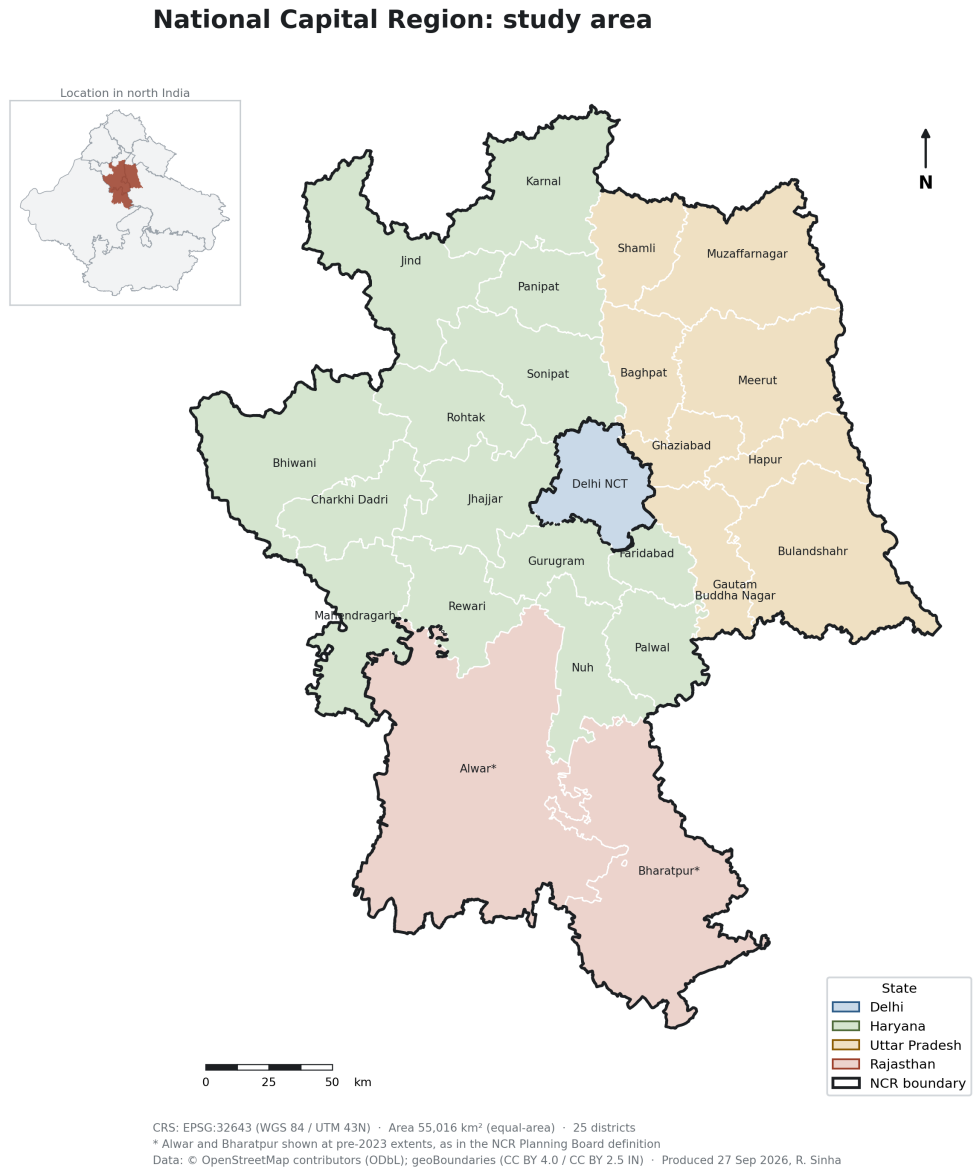


Figure 1: Study area. Alwar and Bharatpur at their pre-2023 extents. EPSG:32643.

2.1 Boundary construction and verification

OpenStreetMap carries Rajasthan's 2023 district reorganisation, in which Alwar and Bharatpur lost territory to Khairthal-Tijara, Kotputli-Behror and Deeg. The NCR is defined on the pre-2023 districts, so they were rebuilt: old Bharatpur = Bharatpur + Deeg; old Alwar = Alwar + Khairthal-Tijara + the part of Kotputli-Behror inside the geoBoundaries ADM2 Alwar polygon. geoBoundaries was rejected for Delhi in an earlier project because its delineation predates 2012; that same property makes it correct here. Slivers under 5 km^2 from the coarser geoBoundaries line were dropped.

Unit	This study (km^2)	Reference (km^2)	Deviation
NCR total	55,016	55,083 (NCRPB)	−0.12%
Alwar, pre-2023	8,355	8,380 (Census 2011)	−0.30%
Bharatpur, pre-2023	5,059	5,066 (Census 2011)	−0.14%

Table 1: Boundary verification. Areas in an India-centred Albers equal-area projection. Reference figures checked 13 October 2026 against the NCRPB constituent-area table (55,083 km^2) and Census 2011 district areas. The two Rajasthan figures sum to 13,446 km^2, which agrees with the NCRPB Rajasthan sub-region (13,447 km^2).

Without the reconstruction, about 3,200 km^2 of Rajasthan would have been omitted from the study area.

3 Data sources

Dataset	Provider / access	Licence	Use	Status
District boundaries	OpenStreetMap, via Overpass API	ODbL	Haryana, UP, Rajasthan districts	Fetched
Delhi NCT boundary	OpenStreetMap, from earlier Delhi project	ODbL	Delhi portion of NCR	Verified
geoBoundaries IND ADM2	William & Mary geoLab	CC BY 4.0	Cut Kotputli-Behror to pre-2023 Alwar	Used
geoBoundaries IND ADM1 (simplified)	William & Mary geoLab; source DataMeet / ECI	CC BY 2.5 IN	Locator inset only	Used
SentinelKilnDB	IIT Gandhinagar Sustainability Lab, Hugging Face	CC BY-NC 4.0	Training and testing the detector	Used
Authors' code	GitHub `rishabh-mondal/NeurIPS_2025`	Not stated	Read to match image processing	Read
Sentinel-2 L2A (harmonised)	Copernicus / ESA, via Google Earth Engine	Copernicus free licence	Median composites for detection; per-scene SWIR for firing	Used
ESA WorldCover 2021 (10 m)	ESA, via Google Earth Engine	CC BY 4.0	Land-cover filter; school built-up check	Used
School locations	NIC SchoolGIS, via `datameet/udise_schools`	Not stated	School points (36,738 after location checks)	Used, checked
School enrolment	UDISE+ (Ministry of Education)	GODL-India (open tables)	Children per school	Not used: access refused (HTTP 403)
WorldPop 2020 age–sex (100 m)	WorldPop, University of Southampton, via Earth Engine	CC BY 4.0	Children aged 5–14	Used (modelled)
OpenStreetMap features	OSM contributors, via Overpass API	ODbL	Non-kiln heat sources (airports, power plants, works)	Used

Dataset	Provider / access	Licence	Use	Status
Google Earth historical imagery	Google / Maxar, Airbus	Viewing only	Visual audit of 25 kilns	Used (viewed, not redistributed)
GRAP orders and schedules	CAQM document index (89 orders, Oct 2022 – May 2026; schedules of 13 Dec 2024 and 21 Nov 2025)	Government publication	Which activities close at each stage	Read
Siting rules	CPCB notification (2022)	Government publication	800 m / 1 km / 500 m rules	Read
Kiln emission factors	World Bank (2020), Table 2.1 (from Weyant et al. 2014 and others)	CC BY 3.0 IGO	$PM_{2.5}$ and BC per kg of brick	Used
Reference areas	NCR Planning Board; Census of India 2011	—	Boundary verification	Verified

4 Methodology

4.1 Training data

SentinelKilnDB contains 114,300 Sentinel-2 patches (128×128 px at 10 m) with 97,648 oriented boxes in three classes: CFCBK (0), FCBK (1), Zigzag (2). Image names give the patch centre as `lat_lon`. 4,076 patches with 5,954 boxes fall inside the NCR; the count includes duplicates from the 30 px patch overlap.

Two properties of the dataset shape the design:

1. **It is not wall-to-wall inside the NCR.** Every NCR patch contains a kiln, so the dataset cannot serve as a complete census. The detector must be run over the full area.

2. **Documentation and code disagree on dates.** The README states November 2023 – February 2024; the download script filters 1 January 2024 – 28 February 2025. This does not affect the change analysis, because both seasons are mapped by the same detector on imagery whose dates this study controls.

4.2 Image processing

Matched to the dataset authors' code: Sentinel-2 surface reflectance bands B4, B3, B2; QA60 cloud mask; patches of 128 px with 30 px overlap; each band of each patch stretched independently to 0–255 by its own minimum and maximum.

4.3 Spatially separated training and test sets

All 4,076 NCR patches form the test set; the detector sees no NCR imagery in training. Training and validation patches come from a 150 km ring around the NCR, split 80/20 by 0.5° spatial blocks. A 2 km strip outside the NCR boundary is excluded, since 1.28 km patches overlap and a border kiln could otherwise appear on both sides. Kiln-free patches (25% of the positive count) are drawn from outside the NCR so the detector learns background.

Set	Patches	With kilns	Kiln boxes
Train (ring)	9,029	7,147	9,573
Validation (ring)	2,657	2,202	2,727
Test (NCR)	4,076	4,076	5,954

Table 3: Training split. Box counts include duplicates from patch overlap.

Leakage in the benchmark's own split. SentinelKilnDB splits patches randomly by class. Because patches overlap by 30 px, a test patch can share ground with a training patch. Measured with a Chebyshev-distance test on patch centres, **76.6% of the benchmark's test patches overlap at least one training patch**. Published scores on that split are therefore likely to be optimistic; the spatially held-out NCR test used here is stricter, and lower scores should be expected.

4.4 NCR imagery

The NCR is covered by 630 tiles of 1,024 × 1,024 px (10.24 km) on a 10 m grid in EPSG:32643, each requested with an explicit pixel transform so that a kiln falls on the same pixel in both seasons. Each season is the per-pixel median of Sentinel-2 surface-reflectance scenes with under 20% cloud, QA60 opaque and cirrus pixels masked, over identical months (1 Nov – 1 Mar) in 2023–24 and 2025–26.

Domain check. Six dataset patches were compared with the same ground cut from this study's 2023–24 composite. Geometry agrees exactly. Radiometry does not: the dataset's single-scene patches show bare or yellow fields where the seasonal median shows green winter crops (median pixel correlation 0.60). Kilns look alike in both; the background differs. The cost of this shift is measured directly in Section 6. It cost 0.14 F1. Retraining on patches cut from this study's own composite was tried and rejected (see *Rejected method*). The shortfall was handled instead by the land-cover and firing checks.

4.5 Reference kiln points

Dataset boxes inside the NCR were converted to ground positions from each patch centre and box offset, and boxes of the same class within 60 m were merged. 5,954 boxes reduce to **4,114 kilns** (3,706 Zigzag, 382 FCBK, 26 CFCBK). One kiln appears in up to eight patches: a quarter of NCR patch centres have a neighbour only ∼350 m away rather than the nominal 980 m stride, consistent with the dataset being built from overlapping state-wise tile runs. The merge removes these duplicates.

4.6 Children resident near kilns

School-level enrolment requires registered UDISE+ access, and the open block-level tables on data.gov.in refused automated access (HTTP 403). As a second, independent exposure measure, resident children aged 5–14 are taken from WorldPop 2020 age–sex structured population (100 m). This is a modelled estimate and is reported as such.

4.7 Change between seasons

Change is measured by firing at fixed locations, not by re-detection, so the detector's misses cannot pose as closures. Each 2023–24 firing-verified kiln is re-tested in the 2025–26 season (1 November 2025 – 1 June 2026) at the same place: *continuing* if hot frequency is still ≥ 0.12, otherwise *stopped*. A 2025–26 detection is *new* if it fires in 2025–26, has no 2023–24 detection within 60 m and did not fire there in 2023–24 (`analysis/kiln_change.py`).

5 Results

5.1 Wall-to-wall NCR scan

The ring detector (confidence 0.25) was run over all 630 tiles for each season: 128 px patches, 30 px overlap, per-patch stretch, detections within 50 m merged. It returned 14,395 kilns in 2023–24 and 13,499 in 2025–26 — 3.5 times the 4,114 reference kilns — with 81% labelled FCBK against 90% Zigzag in the reference. Neither figure was accepted without diagnosis.

Diagnosis. Inside reference-covered patches the detector recovers 65% of reference kilns and 67% of its detections match one, consistent with the held-out test. 73% of detections lie outside reference coverage, where the reference cannot judge them. Land cover separates the two groups (WorldCover 2021, sample of 3,000): matched detections sit on bare (68%) or grass/open (26%) ground, the brick yard; unmatched detections mostly on cropland (48%), trees and shrub (11%) and built-up land (9%).

Land-cover filter. A detection is kept if at least 40% of the WorldCover pixels within 40 m are bare or grass/open. The threshold was chosen as the F1 maximum on reference-covered detections; F1 is flat (0.69) from 0.05 to 0.50, so the choice is not sensitive. There, precision rises from 0.67 to 0.74 for a recall loss of 0.68 to 0.65. Across the NCR the filter removes 57% of detections: **6,140 kilns in 2023–24 and 5,758 in 2025–26**. Kilns built on cropland after 2021 may be wrongly dropped.

Verification by firing. A blind visual review of 60 detections was attempted and abandoned: at basemap resolution a non-specialist reviewer could not reliably tell kilns from brick-coloured fields, so that review would have measured the reviewer, not the map. Detections were instead verified with the hot-spot signal validated above, which needs no interpretation. For each point, hot frequency is the share of clear Sentinel-2 scenes (1 November 2023 – 1 June 2024) with $\mathrm{NHI}_{SWIR} > 0$, maximum within 40 m.

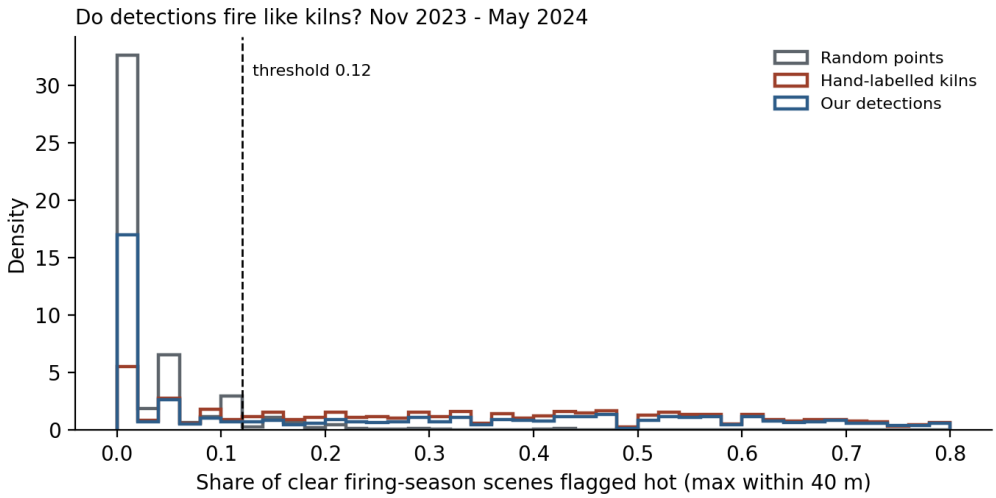


Figure 2: Hot frequency at hand-labelled kilns, random points and detections.

The threshold separating kilns from random points was taken from the data (maximum of Youden's J): hot frequency ≥ 0.12, met by 75.6% of the 4,114 reference kilns and 8.4% of 1,000 random points. 56.1% of filtered detections meet it. If a fraction p of detections are kilns, $56.1 = 75.6p + 8.4(1-p)$, giving an **implied precision of 0.71**. This agrees with the 0.74 measured directly against the reference inside covered patches, from an entirely independent signal. As a further independent check, 78% of the 179 kilns tagged `man_made=kiln` in OpenStreetMap lie within 100 m of a filtered detection.

Two kiln layers are carried forward.

- *All filtered detections* (6,140): precision ≈ 0.71–0.74, recall ≈ 0.65–0.78.
- *Firing-verified kilns* (3,447, hot frequency ≥ 0.12): estimated precision 0.96, since only 8.4% of non-kilns pass the firing test. Kilns idle in 2023–24 are excluded, so this layer is a lower bound on active kilns.

Headline exposure figures use the firing-verified layer and are stated as "at least". The idle-kiln exclusion makes the precision estimates conservative.

Kiln type is not reliable on this imagery. The FCBK/Zigzag split disagrees with the reference by a wide margin, so kiln type is not used in any result until it can be validated.

5.2 Open question: not everything that burns is a kiln

How it arose. Reviewing the firing-verified layer in QGIS, the analyst found a "kiln" on the apron of Indira Gandhi International Airport, Delhi. It passed both checks: the detector saw a small bright rectangle on ground that WorldCover classes as bare, and the firing test saw heat. The firing test asks only *is something hot in shortwave infrared?*; jet exhaust, sun-heated pavement, furnaces, flares and boilers can all answer yes. Firing confirms *a heat source*, not *a kiln*.

How large is the problem? Firing-verified points were overlaid on OpenStreetMap aerodromes, power plants and industrial land (Overpass API, 2,249 polygons; `analysis/heat_source_check.py`). 165 of 3,447 fall inside such a polygon, but most of those polygons are small (median 1.4 ha, 91% under 10 ha) and unnamed — brick fields that mappers drew as industrial land, i.e. real kilns. Counting only points inside a named non-kiln facility, an industrial estate of 10 ha or more, or an airport leaves **17 likely non-kiln heat sources** (15 outside Delhi, 0.4% of the layer): Panipat Refinery, Noida International Airport (under construction), large industrial estates in Gautam Buddha Nagar, Meerut, Hapur, Rewari, Alwar and Gurugram, and two Delhi bus depots.

Status: plausible but unconfirmed, and a lower bound. OpenStreetMap is incomplete — IGI itself was not caught, because the airport is mapped as a multipolygon relation, which the query did not request. The effect on the headline is small (removing the 15 would change the lower estimate by under 0.5% of kilns), but the principle matters: *a thermal signal needs a context check.*

Rectification method. Two independent checks are applied to every firing-verified point, and the headline is reported both before and after the exclusion (`analysis/nonkiln_flag.py`).

1. *Seasonality (a physics check).* Brick kilns shut down in the monsoon (July–September); the go/no-go test found them hot in only 2% of monsoon scenes. Refineries, steel furnaces, power plants and sun-heated airport pavement stay hot all year. A point that is still hot in the monsoon is therefore probably not a kiln. Monsoon hot frequency is measured over July–September 2024 and 2025 (scene cloud < 60%, clouds masked per pixel), and the cut-off is set from the distribution at the 4,114 reference kilns, not assumed (`analysis/monsoon_heat.py`).

2. *Location (a context check).* Points are dropped if they fall inside an airport, power plant, refinery or other works, or a large industrial estate, using OpenStreetMap features mapped either as ways *or as multipolygon relations* — the form in which IGI Airport is mapped and which the first version of the check missed (`analysis/heat_source_check.py`). Small unnamed industrial polygons (< 10 ha) are kept: they are brick fields mapped as industrial land, and behave like kilns (2% monsoon-hot).

Result.

- *Cut-off from the data.* Monsoon hot frequency at the reference kilns has median 0.00 and 99th percentile **0.30**; that percentile is the cut-off, so about 1% of real kilns are sacrificed. Monsoon values are noisier than the go/no-go 2% suggests because few monsoon scenes are clear, which is why a high percentile rather than a small value is used.

- *The two checks agree.* 48% of location-flagged points are monsoon-hot, against 4% of all other points — an independent confirmation that the location check finds non-kilns.
- *They also catch different things.* The IGI apron point has a monsoon hot frequency of 0.00 (jet exhaust is intermittent, not continuous) and is caught only by location; most monsoon-hot points lie in facilities OSM does not map.
- *Removed: 157 of 3,447 points* (location only 12, monsoon only 134, both 11), **leaving 3,290 kilns**. About 38 of the removals are expected to be real kilns (1.09% of the layer). Before removal about 4% of the layer (≈ 148 points) were estimated to be non-kilns; removing ≈ 119 of them raises the estimated precision from 0.96 to about 0.99 (*superseded: the visual audit in Section 6.2 measured 0.91*; plausible, not measured).
- *A further consistency check.* Removed points "stopped" between 2023–24 and 2025–26 only 5.7% of the time, against 13.8% for the kept kilns: year-round heat sources keep burning, as non-kilns should.
- Removals are spread across industrial districts (Alwar, Karnal 12 each; Bhiwani, Rewari, Panipat, Sonipat, Hapur 11 each). Half of Delhi's 18 points (9) are removed by these checks alone.

Duplicate detections. Inspecting the web map, the analyst noticed two kiln points on one kiln. Close pairs (under 120 m) were checked against the hand-labelled reference: pairs matching a single reference kiln (duplicates) lay 50–68 m apart (17 of 21 under 70 m), while pairs matching two reference kilns (genuine neighbours) lay 72–118 m apart (3 of 38 under 70 m). Points closer than 70 m were merged, keeping the one with the highest firing frequency (`analysis/dedupe_kilns.py`): 25 duplicates removed, **leaving 3,265 kilns** (3,256 outside Delhi). Because overlapping 800 m zones cover the same ground, exposure barely changed (605,558 to 605,292 children; schools unchanged at 1,708); kiln counts and H3 were recomputed. The emergency-day test (step 23) and the audit sample were drawn before this step, from the 3,290-point layer; at 0.8% duplicates their conclusions are unaffected.

The removed points are kept, with the reason, in `data/kilns/nonkiln_removed_32643.gpkg`. All results below use the cleaned layer; the values before cleaning are kept in `outputs/tables/*_before_nonkil`

5.3 Schools and children within 800 m of a kiln

800 m zones (the CPCB minimum siting distance) were drawn round 2023–24 kilns. Schools are those passing both location checks (36,738). Children are WorldPop 2020 ages 5–14, summed inside the union of zones so overlapping zones count people once.

Delhi excluded. The 18 firing detections inside Delhi NCT, where brick kilns are not permitted, sit in the Bawana–Narela industrial belt, near Old Delhi, on the Yamuna floodplain and on the IGI Airport apron — more plausibly industrial furnaces, crematoria and other heat sources than kilns. None is near a landfill (closest 2.6 km). The non-kiln checks remove 9; the other 9 are excluded with the rest of Delhi. Including them would add 28,127 children and 27 schools to the lower estimate.

NCR excluding Delhi NCT	Lower: firing-verified	Upper: all filtered
Kilns	3,256	5,914
Kilns with a school within 800 m	34.6%	43.5%
Schools within 800 m of a kiln	1,708	4,542
Children aged 5–14 within 800 m	605,292	1,279,426
Share in Uttar Pradesh districts	68.1%	59.2%
Share in Haryana districts	27.2%	28.7%
Share in Rajasthan districts	4.7%	12.1%
Share in the top 3 districts	36.9%	32.7%
Before non-kiln removal: children	665,738	1,335,400
Before non-kiln removal: schools	1,891	4,712

Table 4: Exposure, 2023–24, after removing non-kiln heat sources. The lower estimate is the headline. Its measured precision is 0.91 (visual audit, 95% CI 0.73–0.98, Section 6.2), so up to about one point in ten may not be a working kiln.

Headline. About 605,000 children aged 5–14 — 7.3% of the NCR's children outside Delhi — live within 800 m of a firing brick kiln, and about 1,708 schools stand within that distance ("about" rather than "at least" because the audited precision is 0.91; Section 6.2). Two-thirds of the exposed children are in the eight Uttar Pradesh districts of the NCR; the top three by number are Muzaffarnagar, Baghpat and Bulandshahr. **By share, Baghpat stands out: 32.8% of its children — one in three — live within 800 m of a firing kiln**, followed by Shamli (16.5%) and Muzaffarnagar (15.3%) (Figure 3). Removing non-kiln heat sources cut the estimate by 9% (60,180 children, 183 schools): industrial sites sit near dense settlement, so each false kiln carried more exposure than a real one.

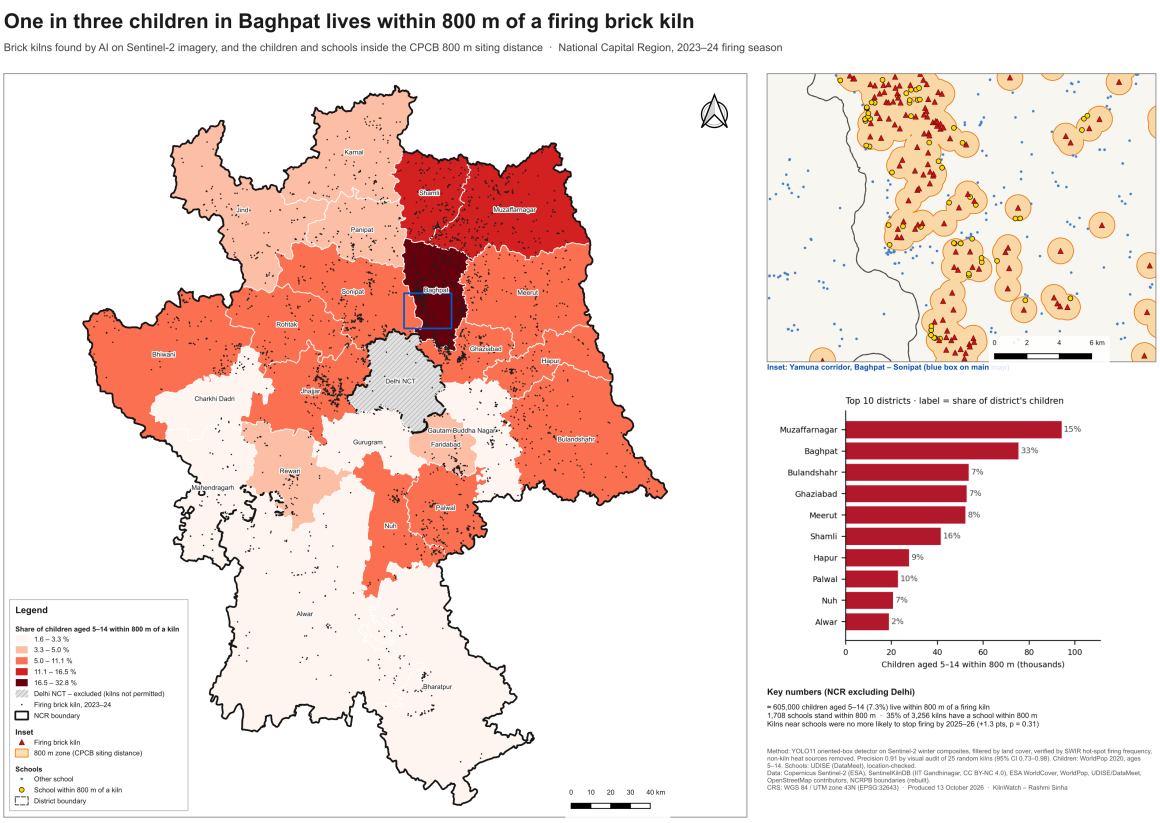


Figure 3: Share of children aged 5–14 living within 800 m of a firing brick kiln, by district, 2023–24 (Jenks classes), with an inset of the Yamuna corridor and the ten districts with most exposed children. Districts are shaded by share rather than count, so that population size is not mistaken for intensity. District child totals: WorldPop 2020 (`analysis/map_exposure_data.py`). EPSG:32643.

Plausibility of WorldPop. WorldPop 2020 gives the NCR 60.4 million people, 18.1% aged 5–14, consistent with the 2011 Census (about 46 million) plus a decade of rapid growth.

H1 — confirmed. Predicted $\geq 25\%$ of kilns with a school within 800 m; measured 34.6% (lower) to 43.4% (upper).

H2 — rejected as stated. Predicted that the top three districts hold more than half the exposed children; they hold 36.9%. Exposure is regional rather than concentrated in a few districts: 68% lies across the eight western Uttar Pradesh districts of the NCR.

5.4 Which kilns stopped firing, 2023–24 to 2025–26

Of 3,447 kilns firing in 2023–24, **2,983 were still firing in 2025–26 and 464 (13.5%) were not**; 71 new firing kilns appeared. After removing non-kiln heat sources and duplicate detections, 453 of 3,265 kilns (13.9%) stopped.

Rejected correction. The plan was to correct the stop rate for the firing test's miss rate: an active reference kiln fails the test 24.4% of the time, so $c = (13.5 - 24.4)/75.6$. That gives a negative rate, which shows the correction is wrong, not that no kiln stopped. The 24.4% comes from *all* reference kilns, including weak and idle ones; the kilns re-tested here were selected *because* they fired strongly (median hot frequency 0.47, four times the threshold), so they are far less likely to drop below it by chance. The miss rate of a selected group cannot be borrowed from the whole population.

What can be said. 13.8% is an *upper bound* on the stop rate of real kilns, since some "stops" are firing-test misses. An earlier estimate here assumed the non-kilns would *inflate* the stop rate; the non-kiln removal showed the opposite (removed points stopped only 5.7% of the

time), so that assumption is withdrawn. The true stop rate is not yet separable from test misses; it needs a test–retest estimate of the firing test on strong firers.

5.5 H3: are kilns near schools closed more often?

If siting rules were enforced around schools, kilns within 800 m of a school would stop more often than other kilns. Delhi NCT excluded (`analysis/h3_stop_near_schools.py`).

2023–24 firing kilns	Kilns	Stopped by 2025–26	Stop rate
School within 800 m	1,127	165	14.6%
No school within 800 m	2,129	283	13.3%
Difference	+1.3 points (95% CI −1.2 to +3.9), χ^2 $p = 0.31$		

Table 5: Stop rates by proximity to a school, non-kiln heat sources removed. Before removal: 14.1% vs 12.9%, +1.2 points ($p = 0.35$). The firing test's error applies equally to both groups, so it shifts both rates, not their difference.

H3 — confirmed. Kilns near schools are no more likely to stop than kilns elsewhere. The confidence interval rules out a difference larger than 3.8 percentage points. The pattern is not uniform by state (Haryana 19.7% vs 15.5%, Uttar Pradesh 12.5% vs 9.9%, Rajasthan 9.9% vs 18.0%, the last on small numbers), but nowhere do near-school kilns close markedly more. Two seasons after the baseline, 962 kilns near schools are still firing.

5.6 Can Sentinel-2 see kilns firing? (Phase 3 go/no-go)

The Normalized Hot-spot Index, $\text{NHI}_{SWIR} = (\text{B12} - \text{B11})/(\text{B12} + \text{B11})$ [3], was computed for every clear Sentinel-2 scene of 2024 at 100 known kilns and 100 control points at least 1.5 km from any known kiln (maximum within 40 m). Hot surfaces emit more at 2.2 μm than 1.6 μm; clay and brick reflect the other way, since clay minerals absorb near 2.2 μm.

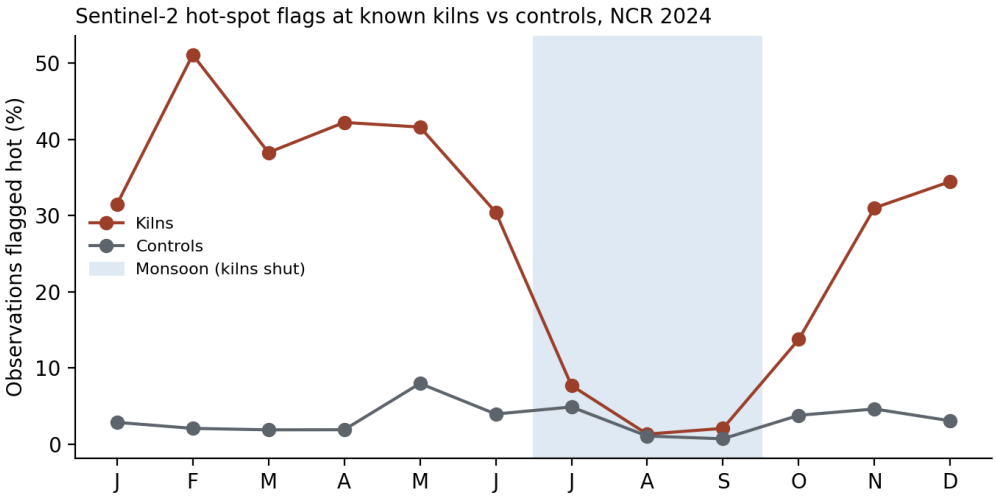


Figure 4: Share of clear observations with NHI > 0, by month, 2024.

	Dry season (Oct–Jun)	Monsoon (Jul–Sep)
Kilns (4,954 obs.)	33.2%	2.0%
Controls (6,451 obs.)	4.0%	1.2%

Table 6: Observations flagged hot. 90% of kilns and 51% of controls were flagged at least once, so a single flag is not evidence; frequency is.

Confound check. The monsoon collapse fits kilns shutting, but also wet ground. A zigzag kiln fires progressively round its trench, so a fire should move; material should not. For the 30 most frequently flagged kilns the hottest 20 m pixel was located on every flagged date: it occupied a median of 9 distinct pixels, with RMS spread 26 m and maximum displacement 84 m — the scale of a kiln loop and well above the ∼10 m multi-date registration error. No kiln's hot spot stayed on one pixel. On its own, a wandering maximum is not conclusive (a uniformly bright target also yields one), but together with the sign of the index and the October onset it is strong evidence that the signal is firing. **Verdict: go.**

5.7 GRAP: do brick kilns have to stop?

All 89 GRAP orders listed by CAQM (October 2022 – May 2026) were read from the Commission's document index, and the Stage III/IV invocations and revocations transcribed with their document identifiers: 40 days with Stage III or IV in force (three windows) in 2023–24, 44 (six) in 2024–25 and 41 (three) in 2025–26.

The GRAP schedules do not close brick kilns. Both schedules CAQM publishes (13 December 2024 and 21 November 2025) were read in full. Kilns appear only under Stage I: pollution rules to be enforced, only approved fuels to be used, and "closure in case of violations". Stage III closes stone crushers and all mining; the "brick / masonry works" it restricts are construction activities, not kilns. Stage IV adds further restrictions and does not mention kilns. Press reports that kilns are "banned at Stage III" do not match the current schedules. The 2023–24 season operated under an earlier schedule no longer listed by CAQM, so kiln coverage then is *[unverified]*; ban-period analysis is restricted to 13 December 2024 onwards.

The question therefore changes from compliance to policy: during the NCR's worst air-quality emergencies, stone crushers and mines are shut but kilns are exempt — do they keep firing? In the 100-kiln test sample (November–February 2024, before the restriction date and so indicative only) kilns were flagged hot in 37.3% of clear observations on Stage III/IV days and 35.2% on other days; controls 4.4% and 2.8%. Controls also rise on those days, the smoggiest, so comparisons must be net of controls.

5.8 The firing calendar, November 2023 – June 2026 (step 22)

For every kiln and every calendar month, the share of clear Sentinel-2 scenes showing heat ($\mathrm{NHI}_{SWIR} > 0$, max within 40 m; scene cloud < 60%, clouds masked per pixel) was measured, with the number of clear scenes, and the same at the 1,000 random control points (`analysis/monthly_firing.py`, `monthly_firing_controls.py`). Months with two or fewer clear scenes per kiln (January fog, monsoon cloud) are flagged as unreliable.

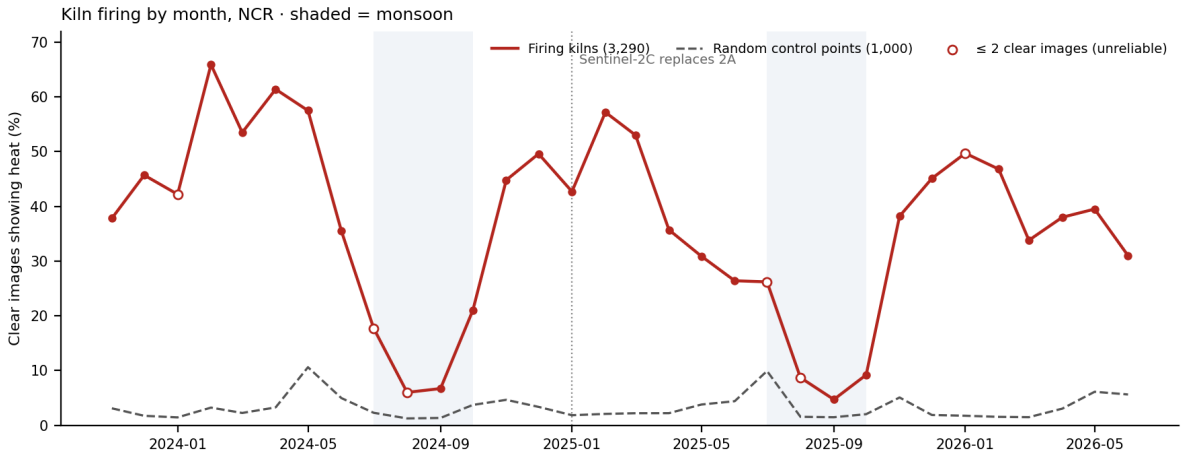


Figure 5: Share of clear images in which kilns (red) and random points (grey) show heat, by month. Hollow points rest on two or fewer clear images.

The season, measured. Kilns relight in October–November, peak in February–May (55–66% of clear images in 2023–24), and drop to near background in August–September (5–7% against 1–2% for controls), matching the known pattern of monsoon shutdown. This is an independent check that the heat signal tracks kiln operation.

A decline in firing, with an observing-system check. Dry-season firing net of the control background fell from 0.50 (2023–24) to 0.42 (2024–25) and 0.37 (2025–26), about a quarter in two seasons. Sentinel-2C replaced Sentinel-2A in January 2025, so a change in the instrument could mimic a change on the ground. The control background stayed flat across that date (dry-season means 4.0%, 3.0% and 3.2%), so the new sensor does not create or suppress false heat. Controls cannot test sensitivity to real fires, however, so the decline is *[plausible, partially checked]*: consistent with the 13.9% stop rate (step 14) plus lower firing among continuing kilns, but not proven to be free of a sensor effect.

5.9 Do kilns keep firing during air-quality emergencies? (step 23)

Design. 13 December 2024 – 28 February 2026, November–February only, 3,290 cleaned kilns and 1,000 random control points. Firing is the share of clear observations with $\mathrm{NHI}_{SWIR} > 0$; the haze effect on smoggy days is removed by subtracting the controls (difference-in-differences).

Rejected first analysis: pseudo-replication. Treating each kiln as an independent observation gave kilns hot on 37.0% of emergency-day scenes against 45.3% on ordinary days, a net effect of −8.2 points with a bootstrap 95% CI of −8.8 to −7.6. Two checks showed this precision was false. (1) Emergency days cluster in November–January, ordinary days include more February, when kilns fire more — a seasonal confound. (2) Re-estimated within each calendar month, the per-kiln effect swung from −45 to +32 points between months. All kilns under one Sentinel-2 pass share that day's atmosphere and viewing conditions, so the effective sample size is the number of *days*, not kilns. The per-kiln interval is withdrawn (`analysis/grap_firing.py`, `grap_firing_bymonth.py`, kept as documented negative results).

Corrected analysis: the day as the unit. For each of 109 days with imagery, net firing = (share of observed kilns hot) − (share of observed controls hot). Days observing fewer than 200 kilns were dropped, and only months containing both emergency and ordinary days were kept, leaving 23 days (10 emergency, 13 ordinary). The emergency effect was estimated within month (month fixed effects by demeaning), with a bootstrap of days within month for the interval and a within-month permutation test (`analysis/grap_firing_daily.py`).

Net firing on emergency days (mean)	42.1 points
Net firing on ordinary days (mean)	51.7 points
Effect of an emergency day, within month	−5.5 points
95% CI (bootstrap of days)	−21.6 to +10.0
Permutation p	0.62

Table 7: Kiln firing on GRAP Stage III/IV days, day-level analysis.

Finding. Kilns clearly *do* fire during the NCR's air-quality emergencies: on emergency days the share of kilns showing heat exceeded the background by 42 points on average. This is consistent with the schedules, which do not require kilns to close. Whether kilns cut back at all could not be determined: the interval spans a substantial reduction and a small increase. With a 5-day revisit and frequent cloud and haze on smog days, Sentinel-2 yields too few usable emergency days (10) to measure a response of plausible size. This is reported as an *inconclusive* test, not as evidence of no effect; a geostationary or daily sensor, or kiln-level monitoring, would be needed.

Lesson recorded. The first result looked precise because of a methodological error, not because the data were informative. Its interval was about 25 times narrower than the honest one.

5.10 How much smoke, and how much of it next to schools? (step 24)

In one line. We multiply how many bricks the kilns bake by how much fine dust each kilogram of brick releases. The result is a range, not a measurement.

Method (`analysis/emissions.py`). One season (2023–24), the 3,256 cleaned kilns outside Delhi:

$$\mathrm{PM}_{2.5}\ (\mathrm{t}) = \text{bricks} \times 3\ \text{kg} \times \mathrm{EF}\ (\text{g/kg fired brick})/10^6.$$

In numbers:

Bricks baked	×	Weight of a brick	×	Dust per kg of brick	=	Dust released
13–20 billion		3 kg		0.12–0.21 g		4,700–12,300 t

Low case: $13.0 \times 10^9 \times 3 \times 0.12/10^6 = 4{,}700$ t. High case: $19.5 \times 10^9 \times 3 \times 0.21/10^6 = 12{,}300$ t.

- *Emission factors* come from World Bank (2020) Table 2.1, averaged from Weyant et al. (2014), Greentech (2012) and others [5, 6]. Zigzag kilns emit 0.12 g $\mathrm{PM}_{2.5}$ and 0.035 g BC per kg. Fixed-chimney kilns emit 0.21 and 0.17. NCR kilns were ordered to convert to zigzag, but the same report warns the zigzag values "seem too optimistic" because most conversions are not built to standard. Zigzag is therefore the *low* case and fixed-chimney the *high* case. We cannot tell the two apart at 10 m (see Future work).

- *Bricks per kiln per season* are **assumed**, not measured: 4 million in the low case (CSE's model kiln, 40 lakh a year) and 6 million in the high case (about 250 billion bricks a year in India divided by about 44,000 kilns gives 5.7 million).

- *Split between kilns.* Bricks cannot be seen from space, but firing can. Each kiln's share of the total is proportional to its 2023–24 firing frequency (`hot_frac`), so busy kilns get more. The NCR total does not depend on this choice; the split between districts and schools does.

Cross-check. Weyant et al.'s South Asian totals (0.19 Tg $\mathrm{PM}_{2.5}$ from 260 billion bricks) imply 0.73 g per brick. Applied to our 13–20 billion bricks, that gives 9,500–14,300 t. This falls inside our range, at the upper end. It shows our numbers are the right order of magnitude, not that they are correct.

	Low (zigzag)	High (fixed chimney)
Bricks fired, season	13.0 bn	19.5 bn
$PM_{2.5}$, all kilns	4,700 t	12,300 t
Black carbon, all kilns	1,400 t	10,000 t
$PM_{2.5}$ from kilns with a school within 800 m	1,500 t	3,900 t
share of total	31.9% (34.6% of kilns)	

Table 8: Kiln emissions, 2023–24 season, NCR excluding Delhi. Rounded; the range reflects assumptions, not a statistical interval.

District	Kilns	Near a school	$PM_{2.5}$ (t, low–high)	of which near schools (low)
Baghpat	412	174	730–1,900	300
Muzaffarnagar	297	70	460–1,200	100
Jhajjar	302	46	440–1,160	65
Sonipat	207	43	380–1,000	73
Ghaziabad	220	83	270–700	93
Shamli	142	66	230–600	116

Table 9: Top six districts by kiln $PM_{2.5}$ (`outputs/tables/emissions_by_district.csv`).

What it shows.

- The NCR's kilns emit roughly **5,000–12,000 t of $PM_{2.5}$** in a season. About **a third** comes from kilns within 800 m of a school.
- Kilns near schools fire slightly *less* than average: 34.6% of kilns produce 31.9% of emissions. This fits H3, which found closures slightly more common near schools.
- **Baghpat** again stands out. It emits the most of any district, and 41% of its kiln emissions (300 of 730 t, low case) come from kilns near schools.
- Most exposed schools have **one** kiln within 800 m. The worst case, JHS Rampur Kheri in Shamli, has **12**, emitting about 25 t of $PM_{2.5}$ a season in the low case. Rampur Kheri, Matiyala (Ghaziabad, 10) and Biharipur (Baghpat, 7) are where on-site air monitoring would tell the most (`outputs/tables/emissions_top_schools.csv`).

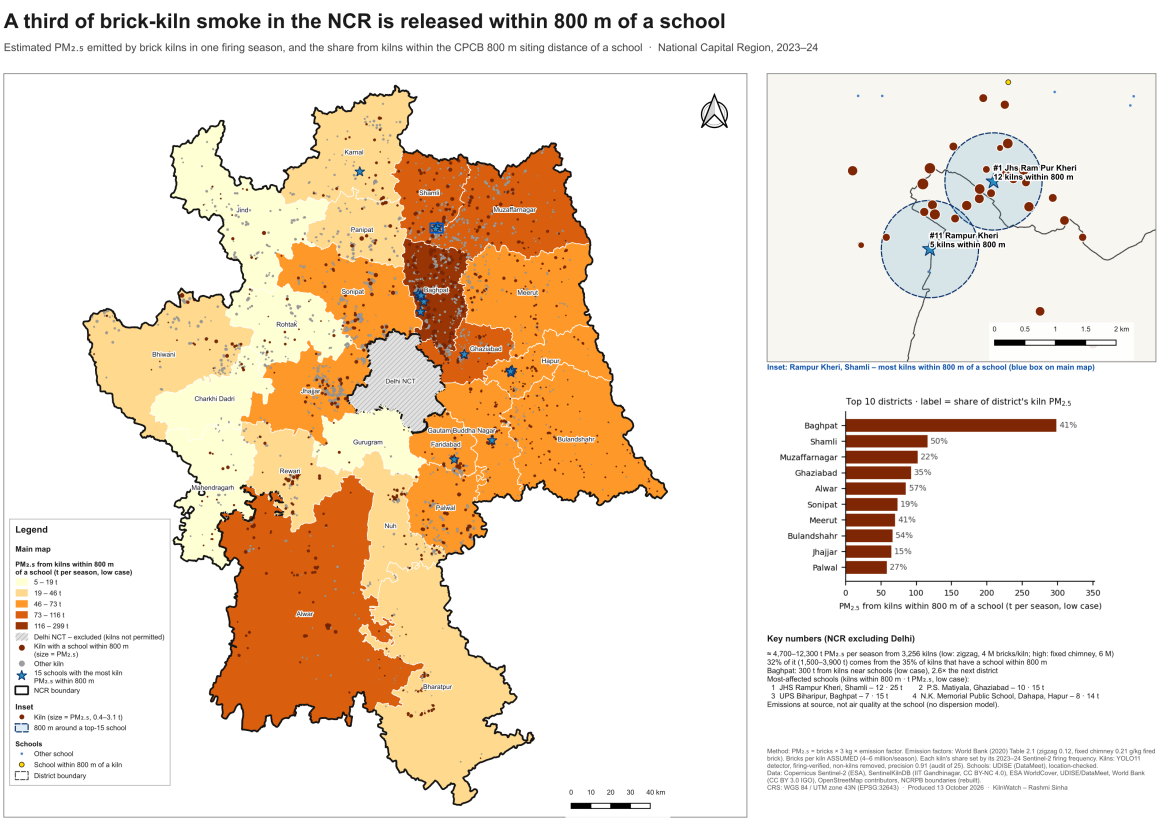


Figure 6: Finished map (step 26): $PM_{2.5}$ from kilns within 800 m of a school, by district (tonnes per season, low case, Jenks classes). Kilns are sized by estimated $PM_{2.5}$. Stars mark the 15 most-affected schools. The inset shows Rampur Kheri, Shamli, where 12 kilns lie within 800 m of one school. Districts are shaded by tonnes rather than tonnes per km^2: these are emissions at source, and dividing by area would imply a spread of smoke that has not been modelled. Large districts with scattered kilns (Alwar) therefore rank higher than a density map would put them. A3 PDF: `outputs/maps/kiln_emissions_map_A3.pdf`; QGIS layout "Emissions map A3".

What it does not show. Emissions are not exposure. Where the smoke goes depends on wind, stack height and mixing depth, which would need a dispersion model (e.g. AERMOD). The table ranks *sources* near schools, not concentrations inside them. Labels: emission factors are *published, measured on few kilns*; bricks per kiln is *assumed*; the firing-frequency split is *measured*.

5.11 Rejected method: fine-tuning on this study's imagery

To remove the domain shift, the ring training patches were re-fetched from this study's 2023–24 median and the detector fine-tuned for five epochs from the best ring model. It made matters worse at every confidence threshold (point-match F1 on this study's imagery, 617 NCR patches):

Confidence threshold	0.05	0.10	0.15	0.25	0.35	0.50
Ring model (kept)	0.58	0.61	0.63	**0.64**	0.62	0.48
Fine-tuned	0.42	0.53	0.54	0.35	0.14	0.01

The likely cause is label noise: dataset labels come from single scenes as late as February 2025, so kilns built after the 2023–24 composite are labelled where the image shows none, teaching the model to under-predict. The fine-tune is retained in the code (`finetune_detector.py`) as a documented negative result. The ring model at confidence 0.25 is used throughout.

The share of Zigzag kilns among NCR patches in the training data (3,669 of 4,234 class occurrences, 87%) is consistent with the directive that kilns in the NCR convert to zigzag technology. This is a plausibility check on the labels, not a finding.

6 Accuracy assessment

6.1 School location accuracy

Of 43,904 NCR school points, 10.5% sit 9–57 km (median) from the district named in their own record and are excluded. Among the rest, a visual check of 30 random points against high-resolution imagery found 14 on a building and 16 on open land (53%, 95% CI roughly 36–70%). Schools often look like the houses around them, so the check asked only the objective question: building or open land.

The same question was automated for every school as the distance to the nearest ESA WorldCover 2021 built-up pixel. All 14 visual "building" points are within 50 m of built-up land; 10 of the 16 "open land" points are too — the point sits on a field or yard beside its village. Agreement 67%.

Distance to nearest built-up	Schools	Share
On built-up	24,371	62.0%
1–50 m	7,476	19.0%
51–100 m	1,825	4.6%
101–300 m	3,066	7.8%
301–800 m	2,136	5.4%
> 800 m	403	1.0%

Table 10: A lower bound on location error, since a school is always a building.

Most points are displaced by less than 300 m, small against an 800 m rule. Schools more than 300 m from any built-up land (6.4%) are treated as unreliable and excluded from the main count; results are reported with and without them.

6.2 Visual audit of 25 random kilns (measured precision)

Why. The precision of the final kiln layer had only been estimated indirectly (from the firing test's error rates). A relict kiln found near Faridabad (next section) showed a failure mode those estimates did not cover, so precision was measured directly.

Method. 25 points drawn at random (seed 2026) from the 3,290-kiln layer, excluding the Faridabad point already examined (`analysis/make_audit_kml.py`). The analyst inspected each in Google Earth Pro historical imagery (Maxar / Airbus). Verdicts used *imagery only* — the satellite heat data were deliberately not consulted, since the audit exists to test them:

- *Active*: kiln structure plus signs of work (drying-brick rows, fresh clay, fired-brick stacks, chimney, change between dates) in an image from the 2023–24 season, *or* working in images on both sides of it ("bracketed", e.g. 2022 and 2025).
- *Relict*: kiln outline but overgrown, no activity. *Not a kiln*: no kiln structure. *Unclear*: anything else.

The rule was tightened during the audit after the analyst noticed that one verdict had been decided with the heat data (circular) and that a 2025-only image cannot exclude a kiln built in 2024; all earlier verdicts were re-checked against it.

Verdict	n	Points
Active (in-season image)	3	1, 19, 22
Active (bracketed)	18	all other active
Relict	1	7
Not a kiln (bare ground)	1	25
Unclear	2	2 (imagery ends 2022), 5 (active 2022, idle 2025)
Precision (unclear excluded)		**21/23 = 0.91** (95% Wilson CI 0.73–0.98)
Worst case (unclear counted wrong)		21/25 = 0.84 (0.65–0.94)

Table 11: Visual audit of the final kiln layer (`outputs/tables/kiln_audit_25.csv`).

Findings.

- *The indirect estimate was too optimistic.* About 0.99 had been estimated from the firing-test statistics; the measured 0.91 places 0.99 outside the 95% interval. The audit, being direct, supersedes it. The two errors were a relict kiln (old outline plus marginal heat) and bare ground that read as hot in 42% of scenes (detector confidence 0.28, just above the 0.25 cut-off).
- *Closures confirmed independently.* For points 5 and 19 the imagery shows the kiln idle or demolished by 2025, and step 14 had independently labelled both *stopped* from heat alone (2025–26 hot frequency 0.06 and 0.03).
- *Position error.* One active kiln (point 22) was found but its point lies about 100 m from the kiln, on the clay yard; negligible against an 800 m zone.
- *Limitations.* 25 points give a wide interval; 18 of 21 active verdicts rest on bracketing, which assumes a kiln working in 2022 and 2025 also worked in 2023–24. One analyst, no second rater.

Implication for the headline. With precision 0.91, roughly one kiln in ten in the lower layer may not be a working kiln, so the "at least" exposure figures may be overstated by up to about 9% on that account; recall below 1 (missed kilns) pushes the other way. The figures are therefore reported as estimates with this stated uncertainty rather than strict lower bounds.

6.3 Spot check: a kiln not visible on the basemap

How it arose. Reviewing the map in QGIS, the analyst found a firing-verified kiln near Faridabad (28.3102 °N, 77.2711 °E) with only fields visible on the Esri World Imagery basemap, and asked how it had been identified.

First reading (later revised, see below). Three tests flagged this spot and it passed all of them in 2023–24, so it probably *was* a kiln then. But it stopped firing, and the current basemap shows no kiln, so it has likely been closed or demolished since.

Test	This point	Typical reference kiln
Detector box / confidence	90 × 41 m, 0.35	∼100 × 40 m; cut-off 0.25
Open land within 40 m	53%	cut-off ≥ 40%
Hot in 2023–24 scenes	33% (7 of 21)	median 33%
Peak NHI_{SWIR}, 2023–24	0.036	median 0.025 (400 kilns)
Hot in monsoon	0%	≈ 0%
2025–26	not detected, 0% hot	— (status: *stopped*)
SentinelKilnDB	none within 2 km	—

Table 12: Evidence for the questioned point (detection 3264).

Why the basemap disagrees. Esri World Imagery is a mosaic of unspecified and generally more recent dates, not the 2023–24 season. The 2023–24 Sentinel-2 median shows a pale, kiln-sized rectangle at this exact spot. A basemap is therefore not a valid reference for a past season.

A hypothesis rejected by measurement. In a SWIR false-colour view of its hottest scene the spot appears white rather than fire-orange, which suggested its heat was too weak to be a real fire. Measuring peak NHI_{SWIR} at 400 reference kilns showed that a kiln's signal at 20 m is always faint (median 0.025, 75th percentile 0.035); this point is stronger than most. The hunch was wrong; the measurement settled it.

Checked against dated imagery — conclusion revised. The analyst inspected Google Earth historical imagery (Maxar / Airbus) for February 2022, April 2023, February 2025 and May 2025. All four show the same thing: a grey oval track about 100 × 40 m — the footprint of a fixed-chimney kiln, matching the detector's 90 × 41 m box — but overgrown, with crops inside, a few huts, and no brick stacks, clay pits, chimney or smoke. No image falls inside the 2023–24 season, but a firing season leaves brick stacks and excavated clay that stay visible for months, and none is present in February 2025.

Verdict: a relict kiln, almost certainly not firing (verified structure, firing rejected). The detector correctly found a kiln *structure*; the firing test wrongly accepted it. Its "hot" readings were marginal (peak 0.036, most below 0.015), and the point sits on a bright concrete building, which may intermittently read NHI_{SWIR} just above zero — plausible, not proven.

Why this matters. This is a failure mode not covered by the non-kiln checks: an *abandoned kiln outline plus marginal heat*. It is not removed by location (no facility) or seasonality (no monsoon heat). Because real kilns are also faint at 20 m (median peak 0.025), a simple heat-strength cut-off would also remove many real kilns, so how common relict kilns are must be measured before any correction. The dated-imagery audit of 25 random kilns (Section 6.2) found 2 non-kilns in 23 decided points: one relict kiln and one patch of bare ground. The earlier reading that the basemap was simply newer than the kiln was wrong: the analyst's check of dated imagery overturned it.

6.4 Independent check of the exposure code (QGIS)

The analyst repeated the exposure count for Baghpat by hand in QGIS 3.40 Processing, without seeing the code's values: *Extract by attribute* (district), *Select by location* (kilns within Baghpat) and *Select within distance* (800 m to schools; schools in Baghpat to kilns anywhere, so kilns across the district border count, as in the code).

Baghpat, cleaned kiln layer	Python	QGIS
Firing kilns in the district	415	415
Kilns with a school within 800 m	174	174
Schools within 800 m of any kiln	263	263

Table 13: Exact agreement between two independent implementations (GeoPandas / SciPy k-d tree vs QGIS Processing).

All three counts agree exactly, confirming the district assignment, the distance test and the treatment of cross-border kilns.

6.5 Detector on held-out NCR patches

YOLO11n-OBB, fine-tuned 13 epochs on the ring set (training stopped once validation mAP50 plateaued at 0.78), was scored on NCR patches it never saw.

Standard metrics, 4,076 NCR patches, dataset imagery: precision 0.31, recall 0.72, mAP50 0.42, mAP50–95 0.27. Recall holds from validation; box precision falls. IoU-based matching is harsh for objects of roughly 10×5 pixels, where a one-pixel offset can drop overlap below 0.5.

Point-matching metrics, which match how the detector is used here (counting kilns): a detection is correct if its centre is within 50 m of a reference kiln, scored in the central 1,000 m of each patch.

Imagery (617 NCR patches)	TP	FP	FN	Precision	Recall	F1
A. Dataset single-scene imagery	598	122	211	0.83	0.74	0.78
B. This study's 2023–24 median	505	277	304	0.65	0.62	0.64

Table 14: Domain shift. Same ground, same detector, same scoring; only the imagery differs. 183 of 800 sampled patches were skipped because the window crossed a tile edge.

The detector is sound on imagery like its training data. Moving it to the seasonal median costs 0.14 F1 — too much to ignore, since every count in this study is made on the median composite. The fix adopted: re-fetch every ring training patch from this study's own composite (labels carry over because kilns do not move) and fine-tune the current model on it. The NCR remains a pure test area. The fine-tune failed; see the rejected-method section. Accuracy of the final kiln layer is therefore measured on the finished product, by the visual audit (precision 0.91, Section 6.2).

7 Limitations

- *Precision.* Measured at 0.91 (95% CI 0.73–0.98) from 25 randomly audited kilns: a small sample, one rater, and 18 of 21 "active" verdicts rest on bracketing (working in 2022 and 2025) rather than in-season imagery.
- *Relict kilns.* Abandoned kiln outlines with marginal heat readings pass both the detector and the firing test; no automatic filter yet removes them.
- *Recall.* At 10 m a kiln spans roughly 10–15 pixels; the detector's recall on this study's imagery is about 0.65–0.78, and the firing test keeps 76% of active kilns, so some working kilns are missing from the lower layer.
- *Children, not pupils.* Exposure counts resident children (WorldPop 2020, modelled) within 800 m, not enrolment at the affected schools, which needs registered UDISE+ microdata.
- *School locations.* 10.5% of UDISE coordinates fall outside their own district and were excluded; the remainder were further filtered by proximity to built-up land (Section 6).
- *Kiln type.* FCBK versus zig-zag classification is unreliable on this imagery and is not used.
- *Emergency-day test.* Inconclusive for lack of clear Sentinel-2 emergency days (10 usable); it shows that kilns fire during emergencies, not whether they respond to them.
- *Emissions estimate.* Bricks per kiln is assumed (4–6 million), and the kiln type (zigzag or fixed chimney) is unknown, so the result is a 2.6-fold range. It is source strength, not air quality at schools.
- *One baseline season.* Exposure is for 2023–24; change is measured only to 2025–26.

Future work

Planned steps moved out of scope (13 October 2026) because they need several days of continuous processing on the available laptop, or a method not yet validated:

- *Kiln maps for 2019, 2021, 2023, 2025* with the same detector, to turn one snapshot into a trend.
- *Zig-zag conversion by district*, testing whether the 2017 order to convert was carried out; requires a kiln-type method validated on 10 m imagery or very-high-resolution imagery.
- *Comparison with state pollution-control-board kiln registers*, to identify kilns seen from space but absent from registers; no usable public register has been located yet.
- *A relict-kiln filter*, e.g. requiring evidence of activity (fresh excavation, change between seasons) in addition to heat, calibrated on an enlarged audit sample.

References

[1] Mondal, R. et al. (2025). *SentinelKilnDB: A Large-Scale Dataset and Benchmark for OBB Brick Kiln Detection in South Asia Using Satellite Imagery.* NeurIPS Datasets and Benchmarks Track.

[2] *Space to Policy: Scalable Brick Kiln Detection and Automatic Compliance Monitoring with Geospatial Data.* arXiv:2412.04065 (2024).

[3] Marchese, F. et al. (2019). A Multi-Channel Algorithm for Mapping Volcanic Thermal Anomalies by Means of Sentinel-2 MSI and Landsat-8 OLI Data. *Remote Sensing*, 11(23), 2876.

[4] Central Pollution Control Board. *Environmental Standards for Brick Kilns.* Gazette notification (2022).

[5] World Bank (2020). *Dirty Stacks, High Stakes: An Overview of Brick Sector in South Asia.* Washington, DC. Table 2.1.

[6] Weyant, C. et al. (2014). Emissions from South Asian Brick Production. *Environmental Science & Technology*, 48(11), 6477–6483.

A Processing parameters

Parameter	Value
Working CRS	EPSG:32643 (UTM 43N)
Area CRS	Albers equal-area, lat_1 12°, lat_2 32°, lat_0 22°, lon_0 80°
Boundary sliver threshold	5 km^2
Patch size / overlap	128 px / 30 px
Bands	B4, B3, B2 (surface reflectance)
Stretch	Per-patch, per-band min–max to 0–255
Siting distances	800 m (schools, habitation, hospitals), 1 km (kilns), 500 m (rivers)
Detector	YOLO11n-OBB, 13 epochs on the ring set; confidence 0.25; detections within 50 m merged
Composites	Per-pixel median, cloud < 20%, QA60 masked, 1 Nov – 1 Mar
Land-cover filter	≥ 40% bare or grass/open WorldCover pixels within 40 m (F1 maximum)
Firing test	$\text{NHI}_{SWIR} = (B12 - B11)/(B12 + B11) > 0$; hot frequency ≥ 0.12 of clear scenes, max within 40 m; season 1 Nov 2023 – 1 Jun 2024
Non-kiln removal	Monsoon (Jul–Sep 2024/25) hot frequency > 0.30 (99th percentile of reference kilns); OSM aerodrome, power plant, works, industrial ≥ 10 ha
Deduplication	Firing kilns < 70 m apart merged, keeping the higher hot frequency
School location check	District label matches location; ≤ 300 m from WorldCover built-up
Children	WorldPop 2020, M_5 + M_10 + F_5 + F_10, 100 m, inside the union of 800 m buffers
Change test	2025–26 season (1 Nov 2025 – 1 Jun 2026), same location and threshold
Emissions	3 kg per brick; 4 / 6 million bricks per kiln; $\text{PM}_{2.5}$ 0.12 / 0.21 g/kg; BC 0.035 / 0.17 g/kg
Map classes	Jenks natural breaks, 5 classes

B Glossary

Term	Meaning
Object detection	A model that locates objects in an image and draws a box around each
OBB	Oriented bounding box: a box rotated to fit the object
YOLO	A family of fast object detection models
Train / validation / test	Data used to learn, to tune, and to give the final score on unseen examples
Held-out	Kept out of training so the test is fair
Precision	Share of detections that are real kilns
Recall	Share of real kilns that were detected
mAP	Mean average precision, a standard detection score from 0 to 1
Benchmark	A published score used for comparison
Fine-tuning	Further training of a pretrained model on new data
Patch	A 128 × 128 px square cut from a larger image (1.28 km across at 10 m)
Min–max stretch	Rescaling brightness so the darkest pixel is 0 and the brightest 255
Surface reflectance	Satellite brightness corrected for atmospheric effects

Term	Meaning
QA60	Sentinel-2 quality band marking cloudy pixels
Sliver	A small spurious polygon where two boundary lines do not coincide
Microdata	Records for individual schools rather than totals
FCBK / CFCBK	Fixed Chimney Bull's Trench Kiln / its circular variant
Zigzag kiln	Cleaner kiln design with zig-zag air flow, mandated in the NCR
CPCB	Central Pollution Control Board
CAQM	Commission for Air Quality Management in NCR and Adjoining Areas
NCRPB	National Capital Region Planning Board
UDISE+	Unified District Information System for Education Plus, India's school database
CC BY-NC	Creative Commons licence: attribution required, non-commercial use only
ODbL	Open Database Licence used by OpenStreetMap
Confound	A factor other than real change that produces apparent change
Pre-registration	Recording hypotheses before analysis
$PM_{2.5}$	Fine particles under 2.5 µm, the most harmful part of smoke
Black carbon (BC)	Soot; part of $PM_{2.5}$ that also warms the climate
t	Tonne (metric): 1,000 kg
Emission factor	Mass of pollutant released per unit of product (here, per kg of fired brick)
Dispersion model	A model of how wind and weather carry pollution from a source

C Decision log

Date	Decision	Reason
26 Sep 2026	Study area = NCR (NCRPB)	Jurisdiction of CAQM
26 Sep 2026	Rebuild pre-2023 Alwar and Bharatpur	NCR is defined on old districts; OSM carries 2023 splits
26 Sep 2026	geoBoundaries only to cut Kotputli-Behror	Pre-2012 delineation, wrong for Delhi, right for old Alwar
26 Sep 2026	Train on SentinelKilnDB	Hand-validated Sentinel-2 benchmark with oriented boxes
27 Sep 2026	Hypotheses H1–H3 recorded	Before any data analysed
27 Sep 2026	Same detector for both seasons	Comparing model output with hand labels would confound method with change
27 Sep 2026	Match authors' per-patch stretch	Read from their code; mismatch would reduce accuracy